

YOKKAICHI, Japan

WMO: 476840

Lat: 34.9400N

Lon: 136.5800E

Elev: 57

StdP: 100.64

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.0	-1.0	-8.7	1.8	3.9	-7.2	2.1	4.0	8.1	6.3	7.2	6.2	1.8	250	0.361

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.4	33.3	25.6	32.0	25.3	30.8	25.0	26.8	30.5	26.3	29.8	25.8	29.1	2.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.0	21.5	28.5	25.4	20.8	28.0	25.0	20.2	27.6	84.8	30.6	82.4	30.0	80.3	29.3	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.7	5.6	4.9	-3.9	35.6	0.9	1.3	-4.5	36.5	-5.0	37.3	-5.5	38.0	-6.1	39.0		
			-4.9	27.9	0.9	0.8	-5.5	28.4	-6.0	28.9	-6.5	29.3	-7.2	29.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	15.5	4.5	5.1	8.3	13.5	18.2	21.9	26.2	27.5	23.7	18.0	12.0	6.9	(6)
		DBStd	8.31	2.17	2.54	2.90	3.12	2.31	2.22	2.26	1.86	2.67	2.77	2.95	2.67	
		HDD10.0	501	172	138	70	7	0	0	0	0	0	13	102		
		HDD18.3	1873	429	370	312	147	30	1	0	1	39	189	355		
		CDD10.0	2523	0	2	17	111	256	357	501	541	411	248	74	6	
		CDD18.3	853	0	0	0	2	27	108	243	283	162	29	0	0	
		CDH23.3	6796	0	0	0	13	135	513	2126	2853	1062	94	0	0	
		CDH26.7	2141	0	0	0	1	16	95	720	1051	252	6	0	0	
(14)	Wind	WSAvg	2.1	2.3	2.4	2.5	2.3	2.1	1.8	1.8	1.9	1.9	1.8	1.8	2.0	(14)
(15)	Precipitation	PrecAvg	1693	57	69	110	137	176	217	186	143	276	157	95	57	(15)
(16)		PrecMax	2187	173	139	217	248	387	322	369	421	559	474	293	122	(16)
(17)		PrecMin	1007	11	27	44	36	81	75	48	28	43	22	7	5	(17)
(18)		PrecStd	296	40	31	43	61	77	68	95	94	139	88	69	30	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	12.9	15.6	19.4	25.2	28.7	31.4	35.1	35.6	32.5	27.5	21.8	16.4		(19)
(20)		0.4% MCWB	8.2	9.8	12.0	15.3	18.9	23.8	25.4	25.9	24.9	21.2	15.9	12.1		(20)
(21)		2% DB	11.4	13.1	17.2	22.4	26.2	29.1	33.1	33.8	30.7	25.7	20.0	14.6		(21)
(22)		2% MCWB	7.2	8.3	10.9	15.0	18.0	22.3	25.7	25.6	24.0	19.6	14.8	10.3		(22)
(23)		5% DB	10.1	11.5	15.4	20.7	24.8	27.5	31.7	32.5	29.3	24.2	18.6	13.1		(23)
(24)		5% MCWB	6.1	7.3	10.2	14.3	17.4	21.6	25.4	25.5	23.6	19.1	14.0	9.2		(24)
(25)		10% DB	8.8	10.1	13.8	19.1	23.3	26.1	30.4	31.4	28.0	22.8	17.2	11.8		(25)
(26)		10% MCWB	5.0	6.4	9.4	13.7	16.9	21.0	25.1	25.3	22.9	18.3	13.3	8.3		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	10.3	12.0	14.6	19.2	21.8	25.6	27.6	27.7	26.4	23.6	17.8	13.7		(27)
(28)		0.4% MCDB	11.3	14.0	16.3	21.0	24.3	29.0	31.7	31.9	29.7	24.9	19.3	14.6		(28)
(29)		2% WB	8.0	9.9	13.1	17.4	20.3	24.3	26.7	27.1	25.6	22.0	16.4	11.4		(29)
(30)		2% MCDB	10.4	11.8	15.2	19.8	23.1	27.2	30.4	30.8	28.2	24.1	18.3	13.3		(30)
(31)		5% WB	6.6	8.2	11.7	16.2	19.4	23.3	26.1	26.6	25.0	20.5	15.3	9.8		(31)
(32)		5% MCDB	9.0	10.4	14.0	18.9	22.2	25.8	29.7	30.1	27.4	22.7	17.4	12.2		(32)
(33)		10% WB	5.5	6.8	10.2	15.0	18.5	22.4	25.6	26.1	24.5	19.2	14.2	8.6		(33)
(34)		10% MCDB	8.3	9.6	13.0	17.7	21.4	24.9	29.0	29.5	26.8	21.7	16.5	11.4		(34)
(35)	Mean Daily Temperature Range	MDBR	7.9	8.6	9.4	9.6	8.9	7.2	6.8	7.4	7.4	8.3	8.7	8.3		(35)
(36)		5% DB MCDBR	9.4	10.6	11.6	11.7	11.6	9.5	8.9	9.2	8.6	9.4	9.8	9.4		(36)
(37)		5% WB MCWBR	6.1	6.8	7.0	6.4	5.8	4.5	3.7	3.6	3.9	4.8	5.9	6.2		(37)
(38)		5% WB MCDBR	8.4	8.7	9.1	8.6	8.5	7.1	7.5	7.5	6.5	7.1	7.6	8.1		(38)
(39)		5% WB MCWBR	6.1	6.6	7.1	5.9	5.3	3.9	3.4	3.2	3.4	4.7	5.6	6.2		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.394	0.433	0.494	0.534	0.542	0.565	0.560	0.532	0.491	0.456	0.415	0.392		(40)
(41)		taud	2.169	2.075	1.930	1.882	1.931	1.945	2.011	2.079	2.162	2.176	2.199	2.191		(41)
(42)		Ebn at Noon	780	792	774	766	766	746	746	760	772	762	753	755		(42)
(43)		Edh at Noon	116	143	179	197	190	187	174	160	141	128	112	108		(43)
(44)	All-Sky Solar Radiation	RadAvg	2.35	3.04	4.00	4.78	5.18	4.66	4.90	5.04	3.98	3.22	2.53	2.12		(44)
(45)		RadStd	0.16	0.24	0.28	0.47	0.57	0.43	0.67	0.45	0.36	0.33	0.23	0.16		(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page